

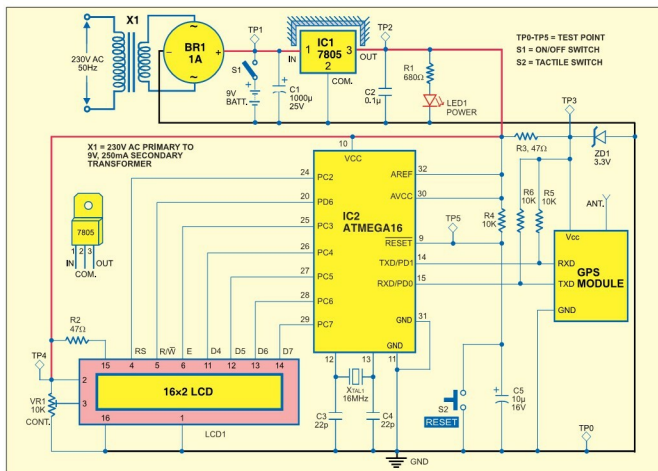


Fig. 2: Circuit of standalone GPS receiver with LCD display

from <http://itouchmap.com/latlong.html>, where you can view the exact location of any point on earth. You can also get the map from http://www.mapsofworld.com/lat_long/india-lat-long.html

You can also use Google Maps for locating the place. It is the common software used for this purpose. You just need to type the latitude and longitude in standard format. For example, enter 28.52, 77.22 in its search box and press 'Search' tab to view the location. It will show you the location somewhere in south Delhi.

Software program

The program for the microcontroller is written in 'C' language using AVR Studio 4.

NMEA protocol. The basic working of the project is based on decoding of National Marine Electronics Association (NMEA) protocol

Test Points

Test point	Details
TP0	0V, circuit ground
TP1	9V supply input
TP2	5V supply for MCU
TP3	3.3V supply for GPS
TP4	5V supply for LCD
TP5	0V when S2 is pressed, otherwise 5V

through software program. The iWave GPS module uses NMEA-0183, which is a subset of NMEA protocol. This protocol includes a set of messages that use ASCII character set and have a defined format. These messages are continuously sent by the GPS module to the interfacing device. Each message starts with a '\$' (hex 0x24) and ends with <CR><LF> (hex 0x0D 0x0A).

These messages include GGA, GGL, GSA, GSV, RMC, VTG and ZDA.

For this project we need not know about all of these messages. We just need the GGA message string, which represents the time, position and fixed data for this application. This string has a format like:

```
$GPGGA, 002153.000, 3342.6618,
N,11751.3858, W, 1.2, 27.0, M, -34.2,
M, 0000*5E
```

Here each field separated by comma (',') represents a particular information. We are using only five of these fields for our purpose of displaying the time, latitude, longitude-direction, longitude and longitude-direction, which are 2nd, 3rd, 5th, 4th and 6th fields, respectively.

The project uses two software programs, namely, 'gps.c' and 'lcd2.h', where 'gps.c' is the main program and 'lcd2.h' driver file for the LCD module.

The main 'C' program starts with the following code lines: